

Scripting Verification Report

Course: Cybersecurity Legislation and Compliance Assessments

Module: Module 8 – Scripted Verification Report (NorthStream)

Student: HASSANI Mohammad Salem

Date: January 29, 2026

Instructor: Asen Grozdanov

Scripted Verification Report (NorthStream)

Introduction and Methodology

This task demonstrates automated verification of security configurations defined in NorthStream's Configuration Baseline (Task 6). The objective is to ensure that documented CIS Benchmark controls are actually enforced on a live system. Verification was performed using OpenSCAP, an industry-standard compliance assessment tool capable of automated PASS/FAIL validation.

Test Environment

The verification was executed on a controlled Ubuntu 24.04 LTS virtual machine running inside VirtualBox. The environment was intentionally configured with selected non-compliant settings to validate detection accuracy.

Results and Analysis

Verification Results Summary

Control 1: Prevent login with empty passwords – FAIL

Control 2: Disable SSH root login – PASS

Control 3: Telnet service disabled (port 23 closed) – PASS

Control 4: Secure permissions on /etc/passwd – PASS

Control 5: System logging enabled – PASS

Failed Control Analysis

The OpenSCAP scan detected the presence of the 'nullok' option in /etc/pam.d/common-auth. This allows authentication without a password, violating CIS benchmarks and creating a critical security gap. This misconfiguration could allow unauthorized system access.

Compliance Integration and Recommendations

Legal and Framework Mapping

This finding impacts compliance with PIPEDA, which requires safeguarding personal information against unauthorized access. From a framework perspective, this maps to NIST CSF PR.AC-4 (Access Permissions) and PR.AC-6 (Least Privilege).

Audit Evidence

The OpenSCAP results provide verifiable, repeatable, and timestamped audit evidence suitable for compliance reviews.

Continuous Monitoring

It is recommended that NorthStream integrate OpenSCAP scans into a weekly compliance check or trigger them after system configuration changes to maintain continuous compliance.